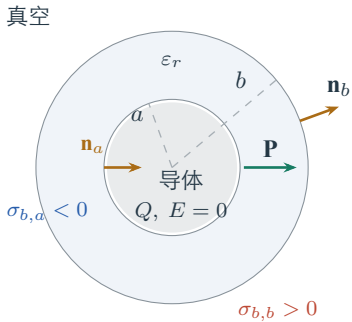
M3-006 介质球壳中的场量、极化电荷与电容



符号示意取 $Q>0,\ \varepsilon_r>1$

$$a<r<b:\ D_r=Q/(4\pi r^2)$$
$$E_r=D_r/(\varepsilon_0\varepsilon_r)$$

$$\mathbf{n}_a=-\mathbf{e}_r,\quad \mathbf{n}_b=\mathbf{e}_r$$
$$q_{b,a}=-Q\alpha,\ q_{b,b}=Q\alpha$$

$$\alpha=1-1/\varepsilon_r$$
介质体内 $\rho_b=0$